

Write-up: The Magic Modbus

5-6 minutes

I downloaded the file and opened it using Wireshark. At the first glance, I saw some Modbus traffic there. I had no idea about Modbus so I just googled it and read about it for a minute.

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I thought of having a look at the conversations in the traffic so I went to *Statistics > Conversations > IPv4*. I found three conversations there:

Ethernet · 1		IPv4 · 3		IPv6	TCP · 3		UDP
Address A ▾	Address B	Packets		Bytes	Packets A → B		Bytes A → B
238.0.0.1	238.0.0.2	44		2,882	22		1,452
238.0.0.3	238.0.0.4	116		7,598	58		3,828
238.0.0.5	238.0.0.6	68		4,454	34		2,244

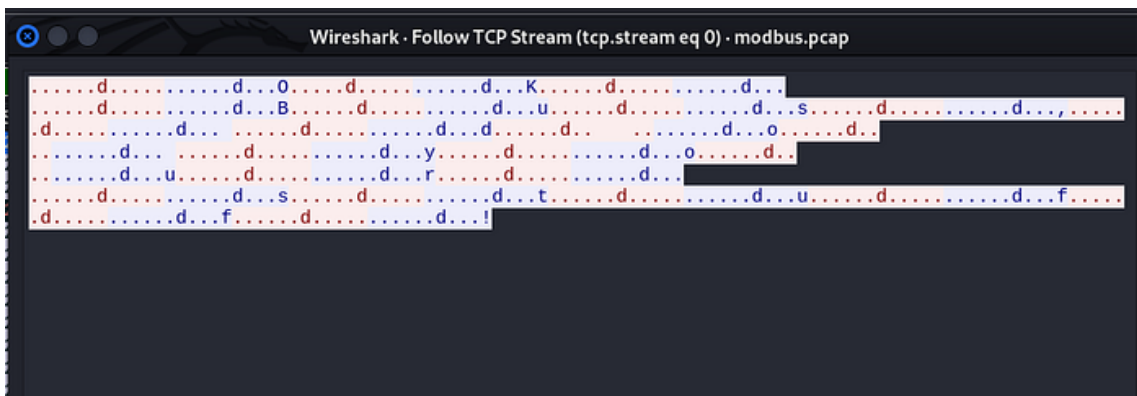


On scrolling through the traffic, I found that there is not much to analyze (only 228 packets). So definitely the message is hidden somewhere between them only.

I opened the TCP stream for the first packet (Ctrl+Alt+Shift+T). I was hoping to find something useful but I didn't noticed anything interesting.

P.S. I later realized that there are some letters in between the stream which make a complete word and if I would have noticed that earlier, it would have been much easier to extract flag ;)

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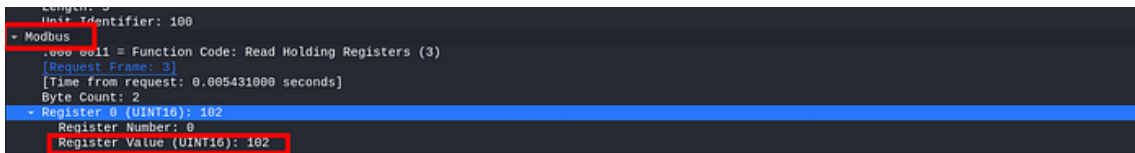


TCP stream

Then, I decided to have a look at the packet details itself. I started analyzing the details of the Modbus protocol in the packets and scrolling among packets to see that which values are continuously changing. In the response packets, I noticed that the Register value is changing continuously.

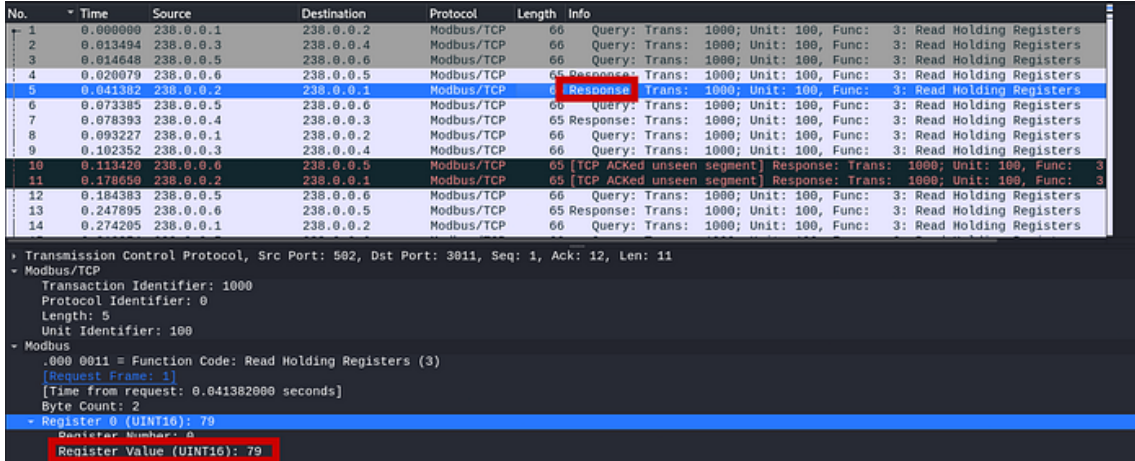
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[illegible]



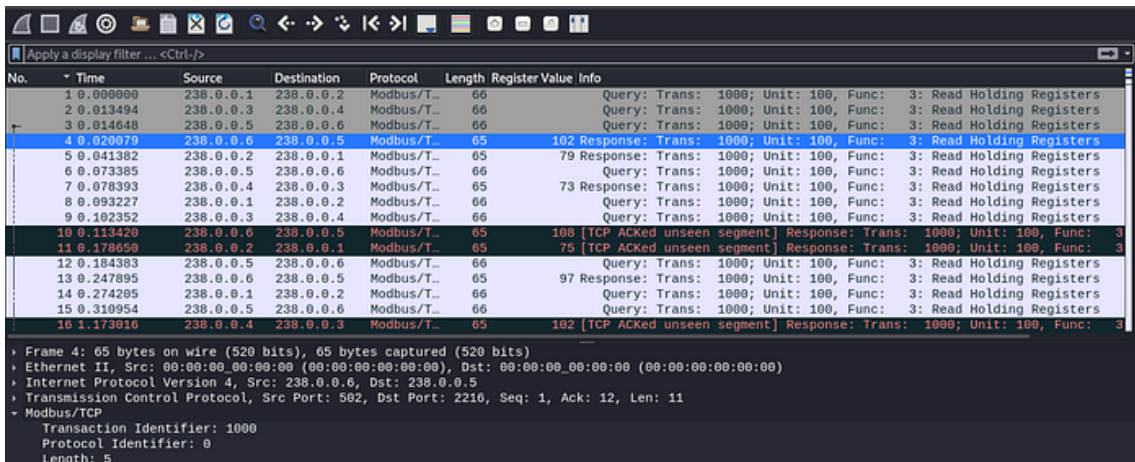
Register Value in Modbus

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These might be some ASCII values in decimal form so I decided to analyze all these register values from Response packets. For this, I right-clicked on the *Register Value* and selected *Apply as Column*. Now I can see all the register values from the main window only.

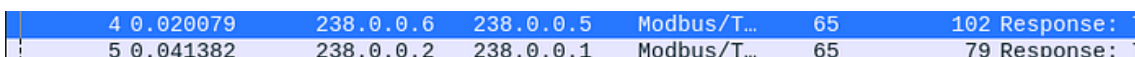
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Register value as column

Another task is to filter for *Response* packets only because there was nothing to analyze in Query packets. I noticed that the *Length* of the Modbus/TCP is 6 for query and 5 for a response. I selected a *Response* packet, right-clicked on the *Length* > *Apply as Filter* > *Selected*. This applied the filter *mbtcp.len == 5*.

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6	0.073385	238.0.0.5	238.0.0.6	Modbus/T...	66	Query: T
7	0.078393	238.0.0.4	238.0.0.3	Modbus/T...	65	Response: T
8	0.093227	238.0.0.1	238.0.0.2	Modbus/T...	66	Query: T
9	0.102352	238.0.0.3	238.0.0.4	Modbus/T...	66	Query: T
10	0.113420	238.0.0.6	238.0.0.5	Modbus/T...	65	108 [TCP ACKed
11	0.178650	238.0.0.2	238.0.0.1	Modbus/T...	65	75 [TCP ACKed
12	0.184383	238.0.0.5	238.0.0.6	Modbus/T...	66	Query: T
13	0.247895	238.0.0.6	238.0.0.5	Modbus/T...	65	97 Response: T
14	0.274205	238.0.0.1	238.0.0.2	Modbus/T...	66	Query: T
15	0.310954	238.0.0.5	238.0.0.6	Modbus/T...	66	Query: T
16	1.173016	238.0.0.4	238.0.0.3	Modbus/T...	65	102 [TCP ACKed

▶ Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
 ▶ Internet Protocol Version 4, Src: 238.0.0.6, Dst: 238.0.0.5
 ▶ Transmission Control Protocol, Src Port: 502, Dst Port: 2216, Seq: 1, Ack: 12, Len:
 ▶ Modbus/TCP
 Transaction Identifier: 1000
 Protocol Identifier: 0
 Length: 5
 Identifier: 100
 ▶ Modbus
 .000 0011 = Function Code: Read Holding Registers (3)
 [Request Frame: 3]
 [Time from request: 0.005431000 seconds]

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mbtcp.len == 5							
No.	Time	Source	Destination	Protocol	Length	Register Value	Info
4	0.020079	238.0.0.6	238.0.0.5	Modbus/T...	65	102	response:
5	0.041382	238.0.0.2	238.0.0.1	Modbus/T...	65	79	response:
7	0.078393	238.0.0.4	238.0.0.3	Modbus/T...	65	73	response:
10	0.113420	238.0.0.6	238.0.0.5	Modbus/T...	65	108	TCP ACKed
11	0.178650	238.0.0.2	238.0.0.1	Modbus/T...	65	75	TCP ACKed
13	0.247895	238.0.0.6	238.0.0.5	Modbus/T...	65	97	response:
16	1.173016	238.0.0.4	238.0.0.3	Modbus/T...	65	102	TCP ACKed
18	1.330725	238.0.0.4	238.0.0.3	Modbus/T...	65	32	response:
19	1.347264	238.0.0.2	238.0.0.1	Modbus/T...	65	32	response:
21	1.372191	238.0.0.6	238.0.0.5	Modbus/T...	65	103	response:
24	1.444236	238.0.0.4	238.0.0.3	Modbus/T...	65	121	response:
26	1.578394	238.0.0.4	238.0.0.3	Modbus/T...	65	111	response:
28	1.724653	238.0.0.4	238.0.0.3	Modbus/T...	65	117	response:
30	2.425264	238.0.0.2	238.0.0.1	Modbus/T...	65	66	response:
32	2.496264	238.0.0.2	238.0.0.1	Modbus/T...	65	117	response:
33	2.505734	238.0.0.6	238.0.0.5	Modbus/T...	65	123	response:

▶ Frame 4: 65 bytes on wire (520 bits), 65 bytes captured (520 bits)
 ▶ Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)

This was looking something promising. Now, this was a difficult task to manually note down all these values thus I decided to use a beautiful command-line-based tool: **Tshark**.

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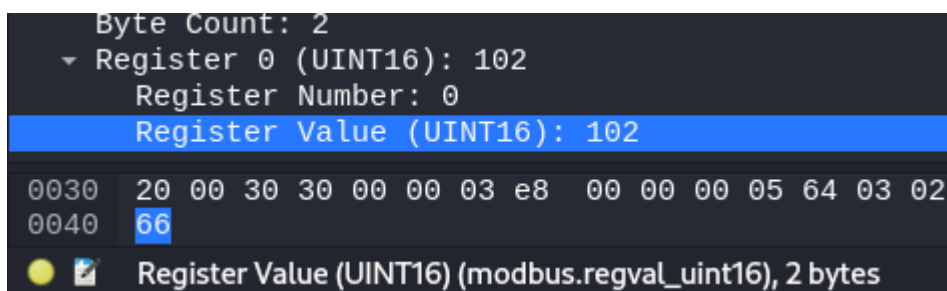
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For extracting Register values using Tshark, we need to get the filter that we need to set and the value that we need to extract from.

We already have the filter **mbtcp.len == 5**. For the register value, right-click on the register value > Copy > Field Name.

Alternatively, the field name can also be seen at the bottom of the screen which is *modbus.regval_uint16*.



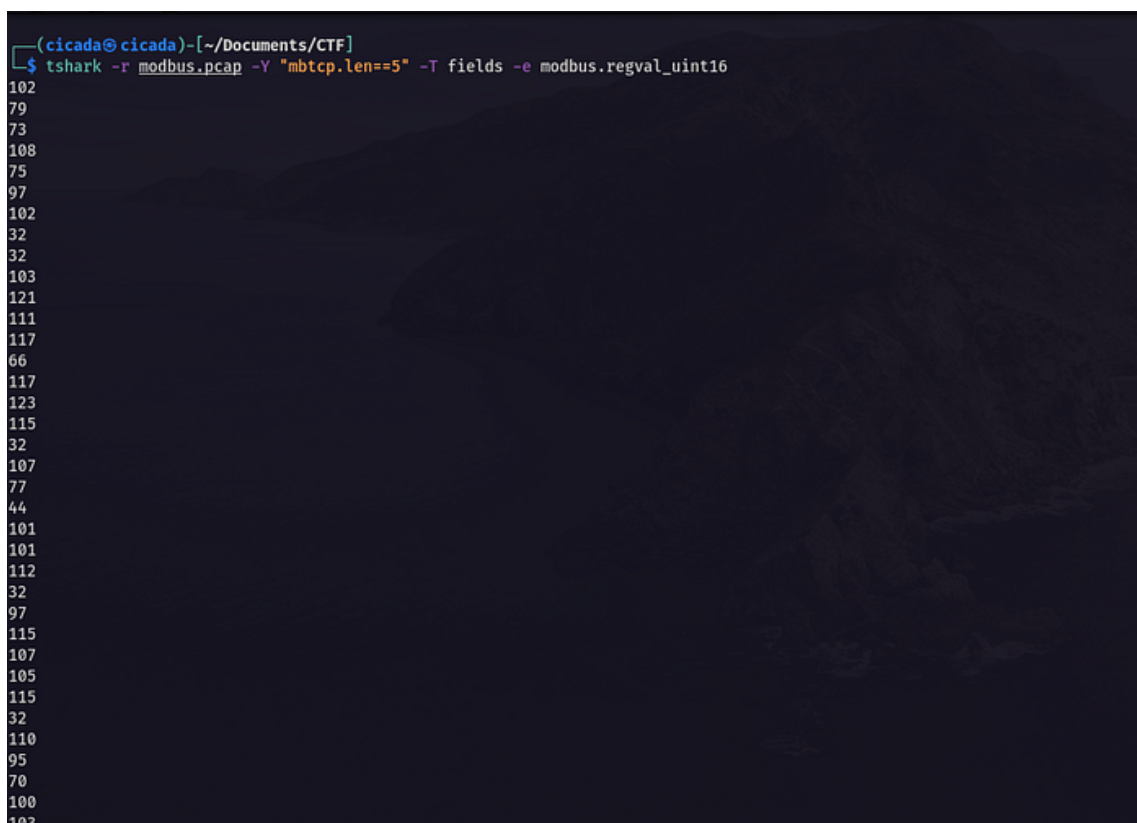
Field name for register value

The command in the terminal that I used was:

```
tshark -r modbus.pcap -Y "mbtcp.len==5" -T fields -e modbus.regval_uint16
```

This gave me all the register values from the Response packets.

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Using Tshark to extract data

Let's save this output and then decode it. To store this output to a text file, I used

```
tshark -r modbus.pcap -Y "mbtcp.len==5" -T fields -e modbus.regval_uint16 > out.txt
```

I created a small python script in the terminal to get the

values:

```
python3 -c "for line in open('out.txt','r'):
print(chr(int(line)),end=' ')"
```

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```
(cicada@cicada)-[~/Documents/CTF]
$ tshark -r modbus.pcap -Y "mbtcp.len==5" -T fields -e modbus.regval_uint16 > out.txt

(cicada@cicada)-[~/Documents/CTF]
$ python3 -c "for line in open('out.txt','r'): print(chr(int(line)),end=' ')"
foIlKaf  gyouBu[s kM,eeP askis n_Fdgro1Z  ZL3qyo_uur sWet0UslfDf!tion_b3s_, syou'll ke0e_p Pr0gUeDtt}ing answers!

(cicada@cicada)-[~/Documents/CTF]
$
```

Hmm.. it doesn't look something like a flag at all? But I can see there are some chunks of words that make sense, like *"you'll"* and *"answers!"*.

This means I must have extracted data in the wrong way. I suddenly recalled that at the starting, I found that there were three conversations going on. Let's try to extract data from these conversation streams individually.

Starting with the first stream:

```
tshark -r modbus.pcap -Y "tcp.stream==0 and
mbtcp.len==5" -T fields -e modbus.regval_uint16
> out.txt
```

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```
(cicada@cicada)-[~/Documents/CTF]
$ tshark -r modbus.pcap -Y "tcp.stream==0 and mbtcp.len==5" -T fields -e modbus.regval_uint16 > out.txt

(cicada@cicada)-[~/Documents/CTF]
$ python3 -c "for line in open('out.txt','r'): print(chr(int(line)),end=' ')"
OK Bus, do your stuff!

(cicada@cicada)-[~/Documents/CTF]
$
```

TCP stream 0

Okay, now this is something interesting. Let's look for the other two streams too.

Similarly, for the second stream:

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```
(cicada@cicada)-[~/Documents/CTF]
$ tshark -r modbus.pcap -Y "tcp.stream==1 and mbtcp.len==5" -T fields -e modbus.regval_uint16 > out.txt

(cicada@cicada)-[~/Documents/CTF]
$ python3 -c "for line in open('out.txt','r'): print(chr(int(line)),end='')"
If you keep asking questions, you'll keep getting answers!

(cicada@cicada)-[~/Documents/CTF]
$
```

TCP stream 1

And finally!! got the flag from the third stream :)

Flag is : flag{Ms_Fr1ZZL3_W0ULD_b3_s0_Pr0UD}

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```
(cicada@cicada)-[~/Documents/CTF]
$ tshark -r modbus.pcap -Y "tcp.stream==2 and mbtcp.len==5" -T fields -e modbus.regval_uint16 > out.txt

(cicada@cicada)-[~/Documents/CTF]
$ python3 -c "for line in open('out.txt','r'): print(chr(int(line)),end='')"
flag{Ms_Fr1ZZL3_W0ULD_b3_s0_Pr0UD}

(cicada@cicada)-[~/Documents/CTF]
$
```

TCP stream 2

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